

Chapter 7: Membrane Structure & Function

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CURRICULUM MODULES INCLUDED

Module 0: Content Introduction (Fluid Mosaic & Fluidity)

Module 2: Passive Transport (Diffusion, Facilitated, Osmosis, Tonicity)

Module 4: Bulk Transport (Endocytosis & Exocytosis)

Module 1: Plasma Membrane (Phospholipids, Proteins, Sugars)

Module 3: Active Transport (Electrochemical Gradients, Na⁺/K⁺ Pump)

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Module 0: Chapter 7 Content Introduction

Source: Chapter 7 Content Introduction_ BSC2010C-26Fall 0001.html

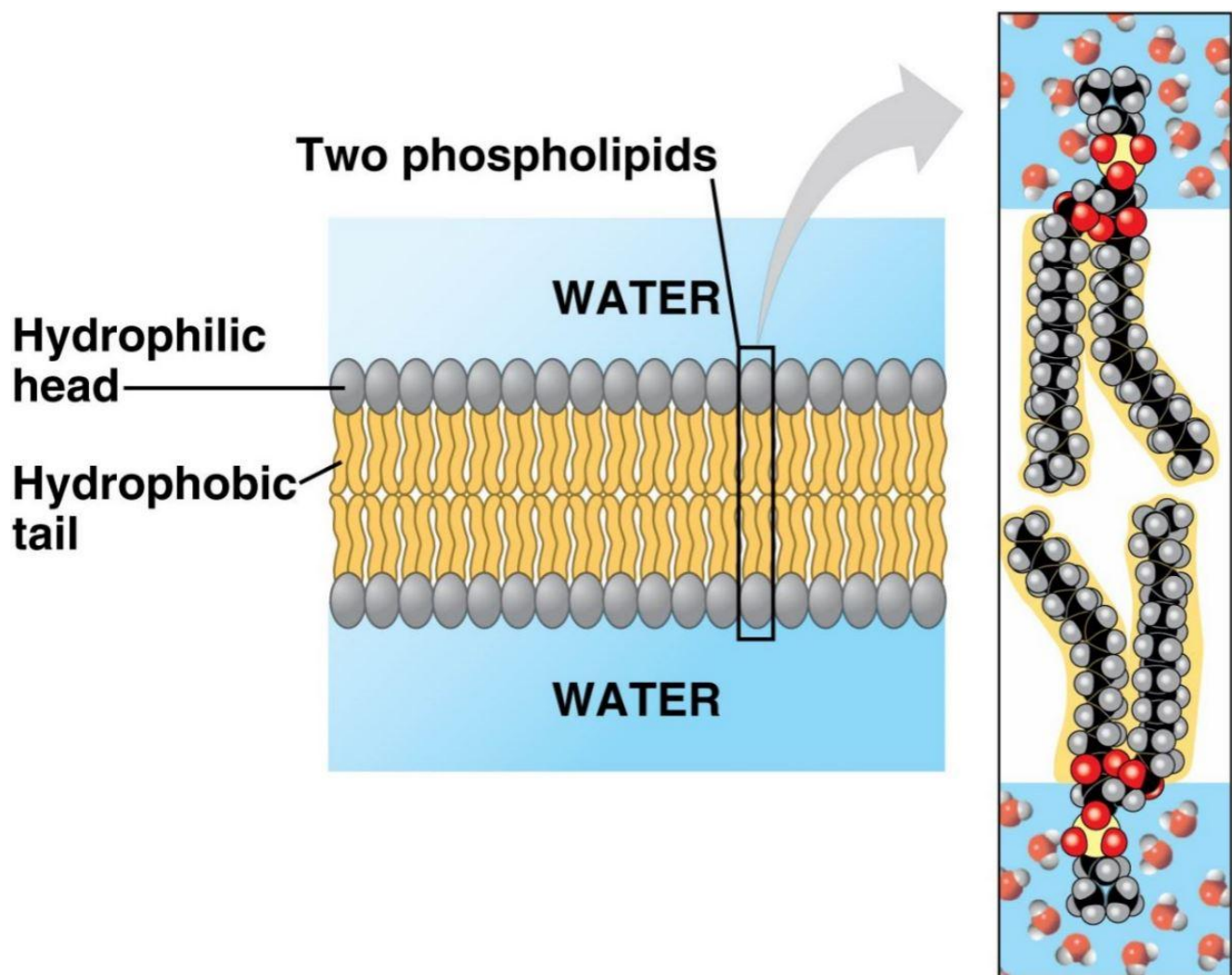
Chapter 7 Content Introduction

Learning Objectives

- Describe how biological membranes are composed of phospholipids
- Explain Fluid Mosaic Model
- Explain how different components of the membrane affect fluidity and permeability
- Identify and describe protein membranes
- Compare various types of membrane diffusion and transport
- Predict water movement based on osmosis

Chapter 7: Membrane Structure and Function

In Chapter 5 we learned about phospholipids, a type of macromolecule composed of a glycerol, two fatty acid tails, a phosphate group, and a small polar molecule at the very top. You should remember that the fatty acid tails are hydrophobic and the heads, with the phosphate group and small polar molecule, are hydrophilic. We also learned that when phospholipids are exposed to water, they organize into a bilayer such that the heads are on the outside and interact with water, and the tails are in between the heads, protected from water.

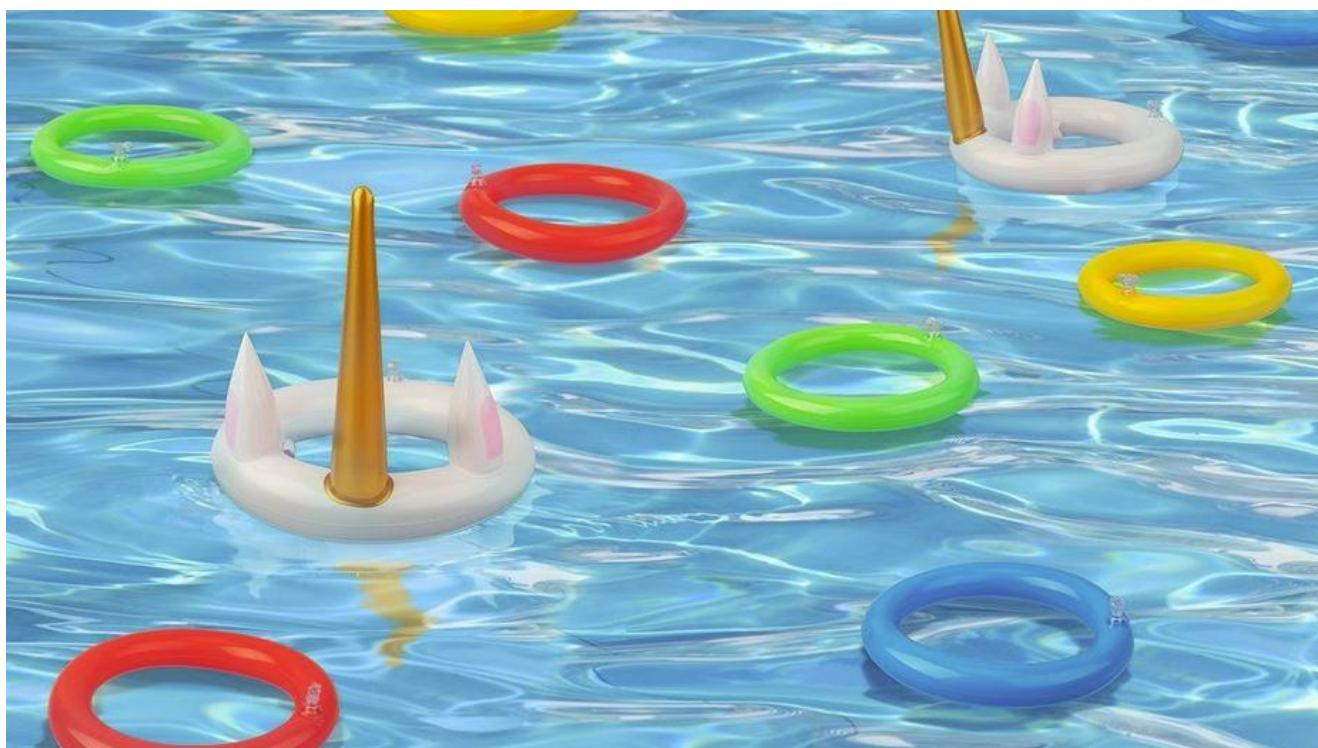


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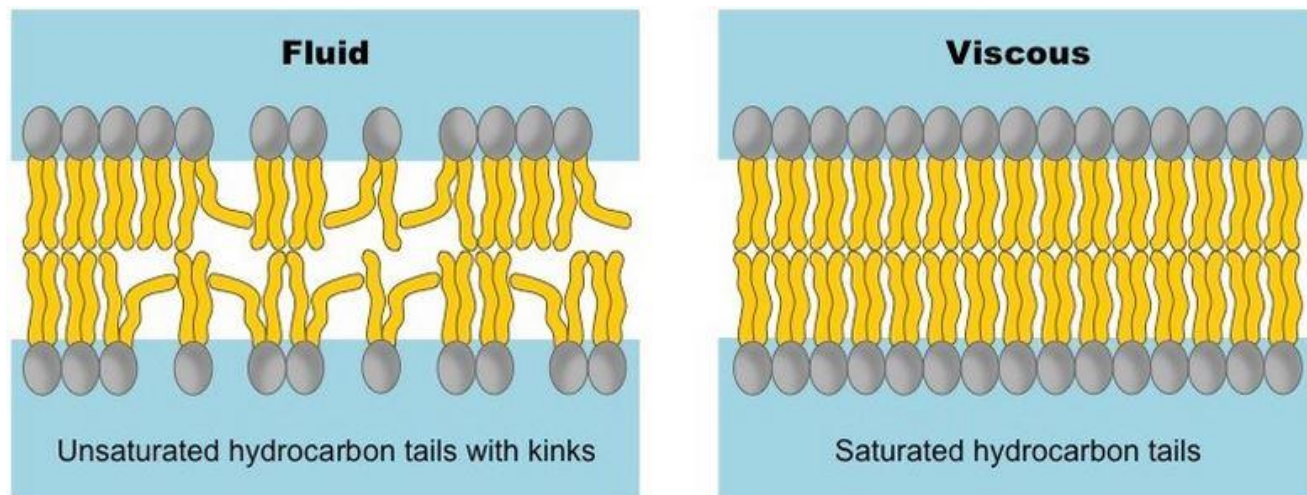
Phospholipid bilayer

The membrane that surrounds a cell is called the **plasma membrane** (or cell membrane) and it is composed of a phospholipid bilayer with proteins and cholesterol molecules embedded within this bilayer. The plasma membrane separates the cell from its external environment and controls the passage of substances into and out of the cell. In this regard, it is **selectively permeable** - some substances are allowed in or out and other substances are not. The Fluid Mosaic Model states that plasma membranes are mosaics of lipids and proteins - the phospholipid bilayer is the "fluid" part, and the proteins are the "mosaic" part.

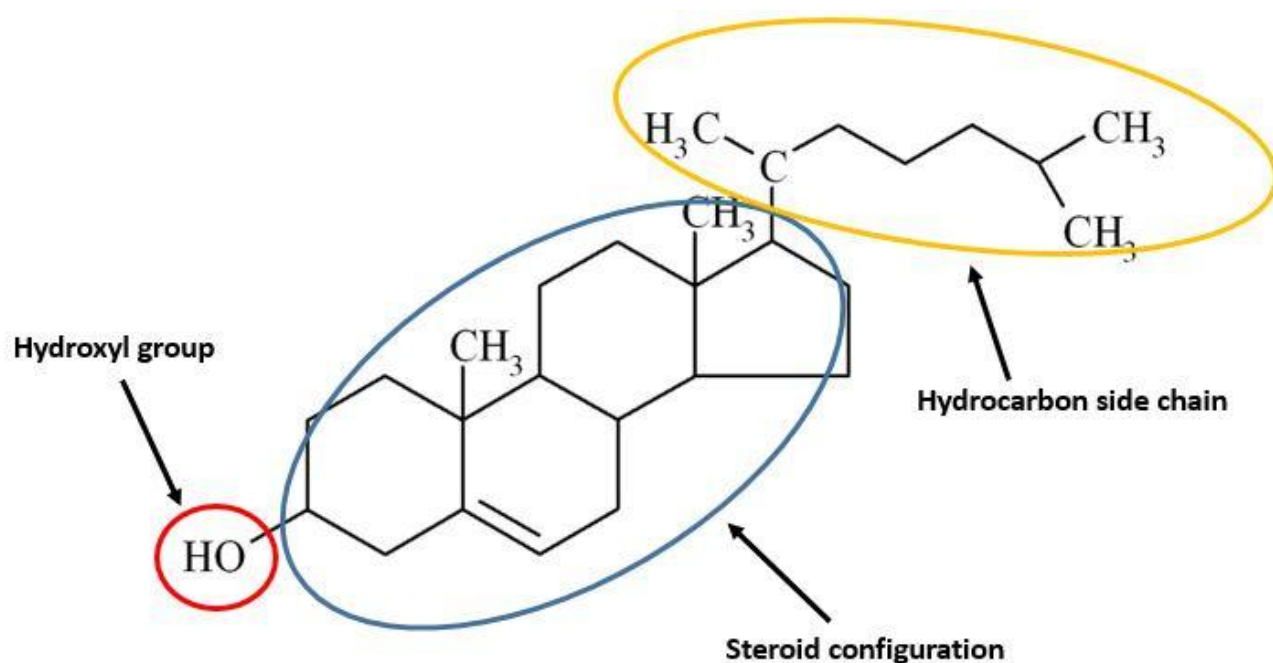
The cell membrane is not static with each phospholipid and the proteins locked in place; it has fluidity so that the phospholipids and proteins can move around. You can picture it a bit like water and toys in a pool. The surface of the water is the phospholipid bilayer and the toys bobbing around in the water are a bit like the proteins - the water and toys are constantly in motion relative to each other.



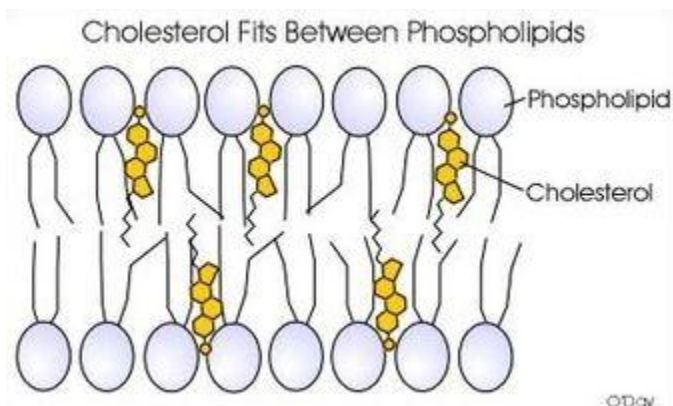
Let's learn a bit more about the phospholipid part of the plasma membrane. You already know that phospholipids have hydrophobic tails and hydrophilic heads; a macromolecule that has both a hydrophobic region and a hydrophilic region is called **amphipathic** (we will use this term a lot in this chapter). Phospholipids are partially responsible for the fluidity of a cell's plasma membrane - how much everything moves around. Each phospholipid has two fatty acid tails; those tails can be saturated or unsaturated. Saturated tails are straight; unsaturated tails have a kink and are slightly bent. Phospholipids with two saturated fatty acid tails can pack in tightly together. Tightly packed phospholipids result in less fluidity of the plasma membrane (it is more viscous).. Phospholipids with one saturated fatty acid tail and one unsaturated fatty acid tail cannot pack tightly together due to the kink in the unsaturated tail. Since these phospholipids cannot pack tightly together, the plasma membrane has more fluidity.



In animal cells, cholesterol also plays a role in membrane fluidity. As you learned in Chapter 5, cholesterol has a typical steroid configuration with the addition of a hydroxyl group on one end and a long hydrocarbon side chain on the other end.



Cholesterol is embedded between the fatty acid tails of the plasma membrane. The hydroxyl group is hydrophilic and is positioned so that it interacts with water. The steroid part of cholesterol and the hydrocarbon side chain are hydrophobic and remain embedded in the fatty acid tails.



Cholesterol helps to maintain plasma membrane fluidity in very cold environments by preventing individual phospholipids from packing together so tightly that the plasma membrane freezes. It also maintains plasma membrane fluidity in very hot environments by holding the individual phospholipids together so they cannot move around so much that the plasma membrane becomes "leaky". To truly understand this, you need to remember that as substances are cooled, molecules pack together tightly until they freeze in position; and when substances are heated up, molecules move further and further apart until they boil away and become a gas. Obviously, either of these would kill the cell and possibly the organism that the cell is within.

The type of fatty acid tails and the abundance of cholesterol are very important in organisms that are cold-blooded, such as fish and reptiles. Arctic fish live in water that is just above the freezing level - membranes with a higher concentration of unsaturated fatty acid phospholipids and a higher concentration of cholesterol helps to prevent the plasma membranes of arctic fish from packing too tightly together and freezing. Bugs and reptiles that live in the desert have to survive temperatures that can exceed 150 degrees Fahrenheit - membranes with a higher

concentration of saturated fatty acid phospholipids and a higher concentration of cholesterol helps to prevent the plasma membranes of these animals from become so fluid that the inside of the cell leaks out.

Interactive Course Widget: Membrane Fluidity widget

CHAPTER READING

Chapter 7 Textbook Reading: [Plasma Membrane](#), [Passive Transport](#), [Active Transport](#), [Bulk Transport](#)

CHAPTER LINKS

Chapter 7 PowerPoint: [Part 1](#); [Part 2](#) ([PDF Part 1](#), [PDF Part 2](#))

[Chapter 7 Post-Quiz](#)

[Chapter 7 Simple Practice Problems](#), [Osmosis Practice Problems](#) ([answers to Osmosis Practice](#))

[Exam 2 Study Guide](#)

Module 1: Chapter 7 Plasma Membrane

Source: Chapter 7 Plasma Membrane_ BSC2010C-26Fall 0001.html

Chapter 7 Plasma Membrane

Learning Objectives

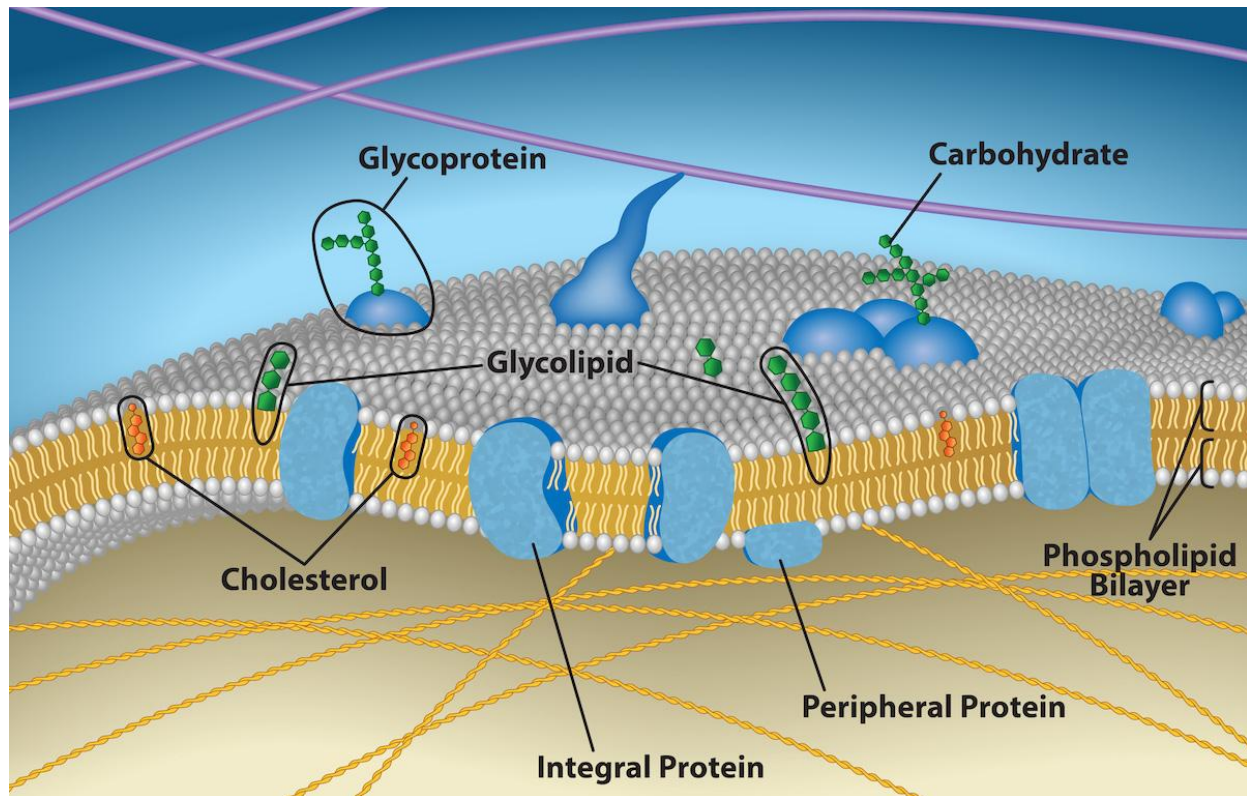
By the end of this section, you will be able to do the following:

- Understand the cell membrane fluid mosaic model
- Describe phospholipid, protein, and carbohydrate functions in membranes
- Discuss membrane fluidity

A cell's plasma membrane defines the cell, outlines its borders, and determines the nature of its interaction with its environment (see [Table 5.1](#) for a summary). Cells exclude some substances, take in others, and excrete still others, all in controlled quantities. The plasma membrane must be very flexible to allow certain cells, such as red and white blood cells, to change shape as they pass through narrow capillaries. These are the more obvious plasma membrane functions. In addition, the plasma membrane's surface carries markers that allow cells to recognize one another, which is vital for tissue and organ formation during early development, and which later plays a role in the immune response's "self" versus "non-self" distinction.

Fluid Mosaic Model

Scientists identified the plasma membrane in the 1890s, and its chemical components in 1915. The principal components they identified were lipids and proteins. The fluid mosaic model describes the plasma membrane structure as a mosaic of components—including phospholipids, cholesterol, proteins, and carbohydrates—that gives the membrane a fluid character. Plasma membranes range from 5 to 10 nm in thickness. For comparison, human red blood cells, visible via light microscopy, are approximately 8 μm wide, or approximately 1,000 times wider than a plasma membrane. The membrane does look a bit like a sandwich ([Figure 5.2](#)).



This illustration shows a phospholipid bilayer with proteins and cholesterol embedded in it. Integral membrane proteins span the entire membrane. Protein channels are integral membrane proteins with a central pore through...

Figure 5.2 The plasma membrane fluid mosaic model describes the plasma membrane as a fluid combination of phospholipids, cholesterol, and proteins. Carbohydrates attached to lipids (glycolipids) and to proteins (glycoproteins) extend from the membrane's outward-facing surface. Credit: Rao, A., Ryan, K., Fletcher, S., Hawkins, A. and Tag, A. Department of Biology, Texas A&M University.

A plasma membrane's principal components are lipids (phospholipids and cholesterol), proteins, and carbohydrates attached to some of the lipids and proteins. A phospholipid is a molecule consisting of glycerol, two fatty acids, and a phosphate-linked head group. Cholesterol, another lipid comprised of four fused carbon rings, is situated alongside the phospholipids in the membrane's core. The protein, lipid, and carbohydrate proportions in the plasma membrane vary with cell type and different cell membranes. Carbohydrates are present only on the plasma membrane's exterior surface and are attached to proteins, forming glycoproteins, or attached to lipids, forming glycolipids.

Phospholipids

The membrane's main fabric comprises phospholipid molecules. The hydrophilic or "water-loving" areas of these molecules are in contact with the aqueous fluid both inside and outside the cell. Hydrophobic, or water-hating molecules, tend to be non-polar. They interact with other non-polar molecules in chemical reactions, but generally do not interact with polar molecules. When placed in water, hydrophobic molecules tend to form a ball or cluster. The phospholipids' hydrophilic regions form hydrogen bonds with water and other polar molecules on both the cell's exterior and interior. Thus, the membrane surfaces that face the cell's interior and exterior are hydrophilic. In contrast, the cell membrane's interior is hydrophobic and will not interact with water. Therefore, phospholipids form an excellent two-layer cell membrane that separates fluid within the cell from the fluid outside the cell.

A phospholipid molecule (Figure 5.3) consists of a three-carbon glycerol backbone with two fatty acid molecules attached to carbons 1 and 2, and a phosphate-containing group attached to the third carbon. This arrangement gives the overall molecule a head area (the phosphate-containing group), which has a polar character or negative charge, and a tail area (the fatty acids), which has no charge. The head can form hydrogen bonds, but the tail cannot. Scientists call a molecule with a positively or negatively charged area and an uncharged, or non-polar, area amphiphilic or “dual-loving.”

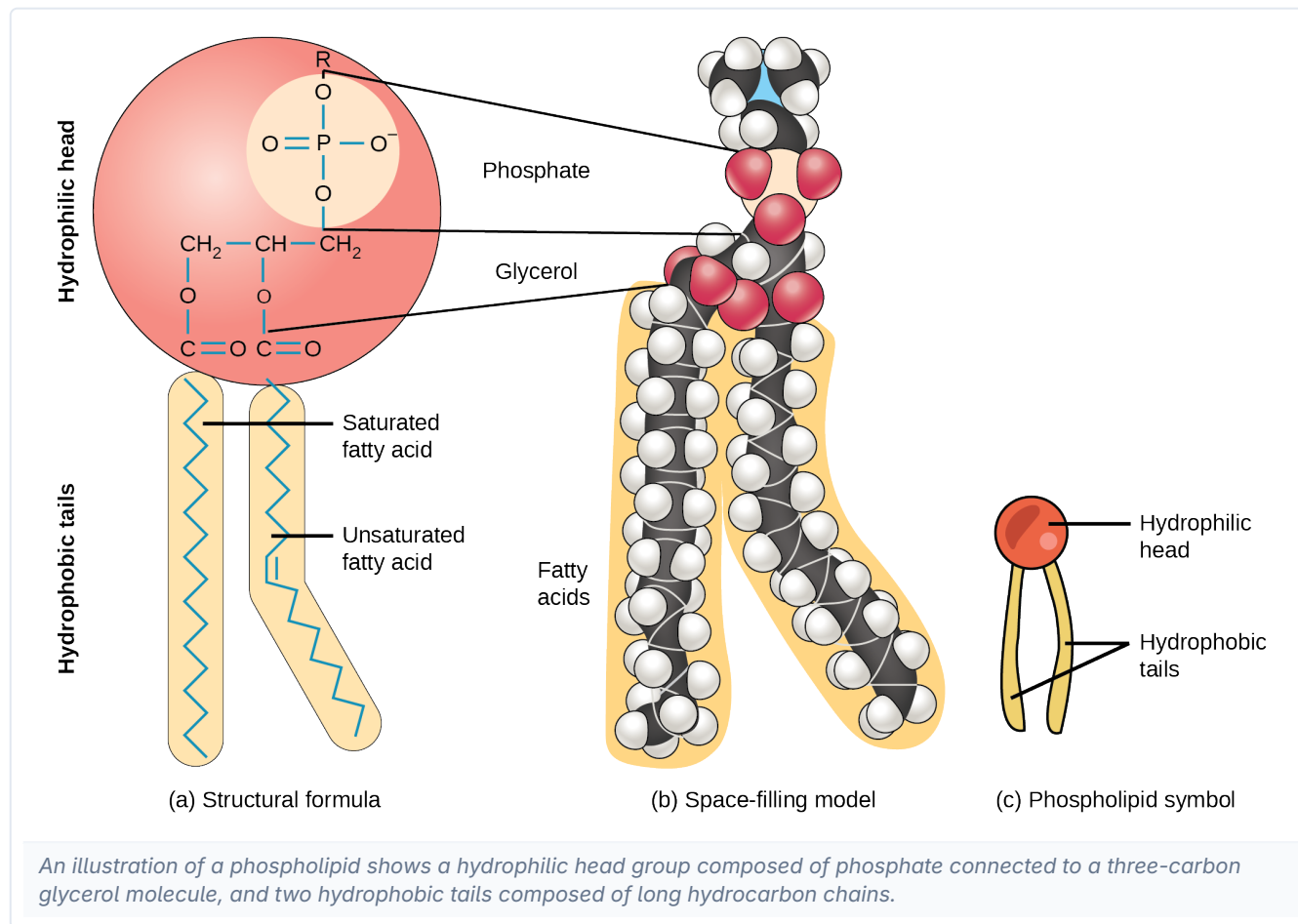
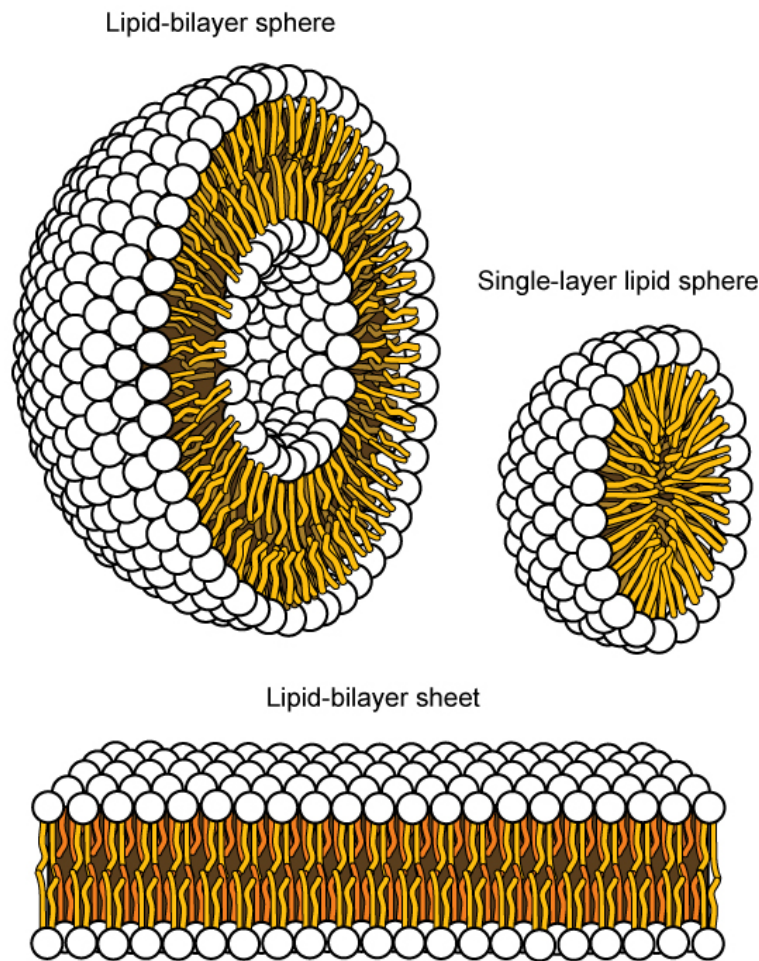


Figure 5.3 A hydrophilic head and two hydrophobic tails comprise this phospholipid molecule. The hydrophilic head group consists of a phosphate-containing group attached to a glycerol molecule. The hydrophobic tails, each containing either a saturated or an unsaturated fatty acid, are long hydrocarbon chains.

This characteristic is vital to the plasma membrane's structure because, in water, phospholipids arrange themselves with their hydrophobic tails facing each other and their hydrophilic heads facing out. In this way, they form a lipid bilayer—a double layered phospholipid barrier that separates the water and other materials on one side from the water and other materials on the other side.



The image on the left shows a spherical lipid bilayer, shown as a half sphere whose surface is covered in the spherical polar heads, and thin, strandlike extend inward. In the core of the sphere is another half sphere, ...

Figure 5.4 In an aqueous solution, phospholipids usually arrange themselves with their polar heads facing outward and their hydrophobic tails facing inward. (credit: modification of work by Mariana Ruiz Villareal)

Proteins

Proteins comprise the plasma membranes' second major component. Integral proteins integrate completely into the membrane structure, and their hydrophobic membrane-spanning regions interact with the phospholipid bilayer's hydrophobic region ([Figure 5.2](#)). Some span only part of the membrane while others stretch from one side to the other, and are exposed on either side. This protein type has a hydrophilic region or regions, and one or several mildly hydrophobic regions. This arrangement of protein regions orients the protein alongside the phospholipids, with the protein's hydrophobic region adjacent to the phospholipids' tails and the protein's hydrophilic region or regions protruding from the membrane and in contact with the cytosol or extracellular fluid.

Peripheral proteins are on the membranes' exterior and interior surfaces, attached either to integral proteins or to phospholipids. Peripheral proteins, along with integral proteins, may serve as enzymes, as structural attachments for the cytoskeleton's fibers, or as part of the cell's recognition sites. Scientists sometimes refer to these as "cell-specific" proteins. The body recognizes its own proteins and attacks foreign proteins associated with invasive pathogens.

Carbohydrates

Carbohydrates are the third major plasma membrane component. They are always on the cells' exterior surface and are bound either to proteins (forming glycoproteins) or to lipids (forming glycolipids) ([Figure 5.2](#)). These carbohydrate chains may consist of 2–60 monosaccharide units and can be either straight or branched. Along with peripheral proteins, carbohydrates form specialized sites on the cell surface that allow cells to recognize each other. These sites have unique patterns that allow for cell recognition, much the way that the facial features unique to each person allow individuals to recognize him or her. This recognition function is very important to cells, as it allows the immune system to differentiate between body cells ("self") and foreign cells or tissues ("non-self"). Similar glycoprotein and glycolipid types are on the surfaces of viruses and may change frequently, preventing immune cells from recognizing and attacking them.

Membrane Fluidity

The membrane's mosaic characteristic helps to illustrate its nature. The integral proteins and lipids exist in the membrane as separate but loosely attached molecules. These resemble the separate, multicolored tiles of a mosaic picture, and they float, moving somewhat with respect to one another. The membrane is not like a balloon, however, that can expand and contract; rather, it is fairly rigid and can burst if penetrated or if a cell takes in too much water. However, because of its mosaic nature, a very fine needle can easily penetrate a plasma membrane without causing it to burst, and the membrane will flow and self-seal when one extracts the needle.

The membrane's mosaic characteristics explain some but not all of its fluidity. There are two other factors that help maintain this fluid characteristic. One factor is the nature of the phospholipids themselves. In their saturated form, the fatty acids in phospholipid tails are saturated with bound hydrogen atoms. There are no double bonds between adjacent carbon atoms. This results in tails that are relatively straight. In contrast, unsaturated fatty acids do not contain a maximal number of hydrogen atoms, but they do contain some double bonds between adjacent carbon atoms. A double bond results in a bend in the carbon string of approximately 30 degrees ([Figure 5.3](#)).

Thus, if decreasing temperatures compress saturated fatty acids with their straight tails, they press in on each other, making a dense and fairly rigid membrane. If unsaturated fatty acids are compressed, the "kinks" in their tails elbow adjacent phospholipid molecules away, maintaining some space between the phospholipid molecules. This "elbow room" helps to maintain fluidity in the membrane at temperatures at which membranes with saturated fatty acid tails in their phospholipids would "freeze" or solidify. The membrane's relative fluidity is particularly important in a cold environment. A cold environment usually compresses membranes comprised largely of saturated fatty acids, making them less fluid and more susceptible to rupturing. Many organisms (fish are one example) are capable of adapting to cold environments by changing the proportion of unsaturated fatty acids in their membranes in response to lower temperature.

Animals have an additional membrane constituent that assists in maintaining fluidity. Cholesterol, which lies alongside the phospholipids in the membrane, tends to dampen temperature effects on the membrane. Thus, this lipid functions as a buffer, preventing lower temperatures from inhibiting fluidity and preventing increased temperatures from increasing fluidity too much. Thus, cholesterol extends, in both directions, the temperature range in which the membrane is appropriately fluid and consequently functional.

Module 2: Chapter 7 Passive Transport

Source: Chapter 7 Passive Transport_ BSC2010C-26Fall 0001.html

Chapter 7 Passive Transport

Learning Objectives

By the end of this section, you will be able to do the following:

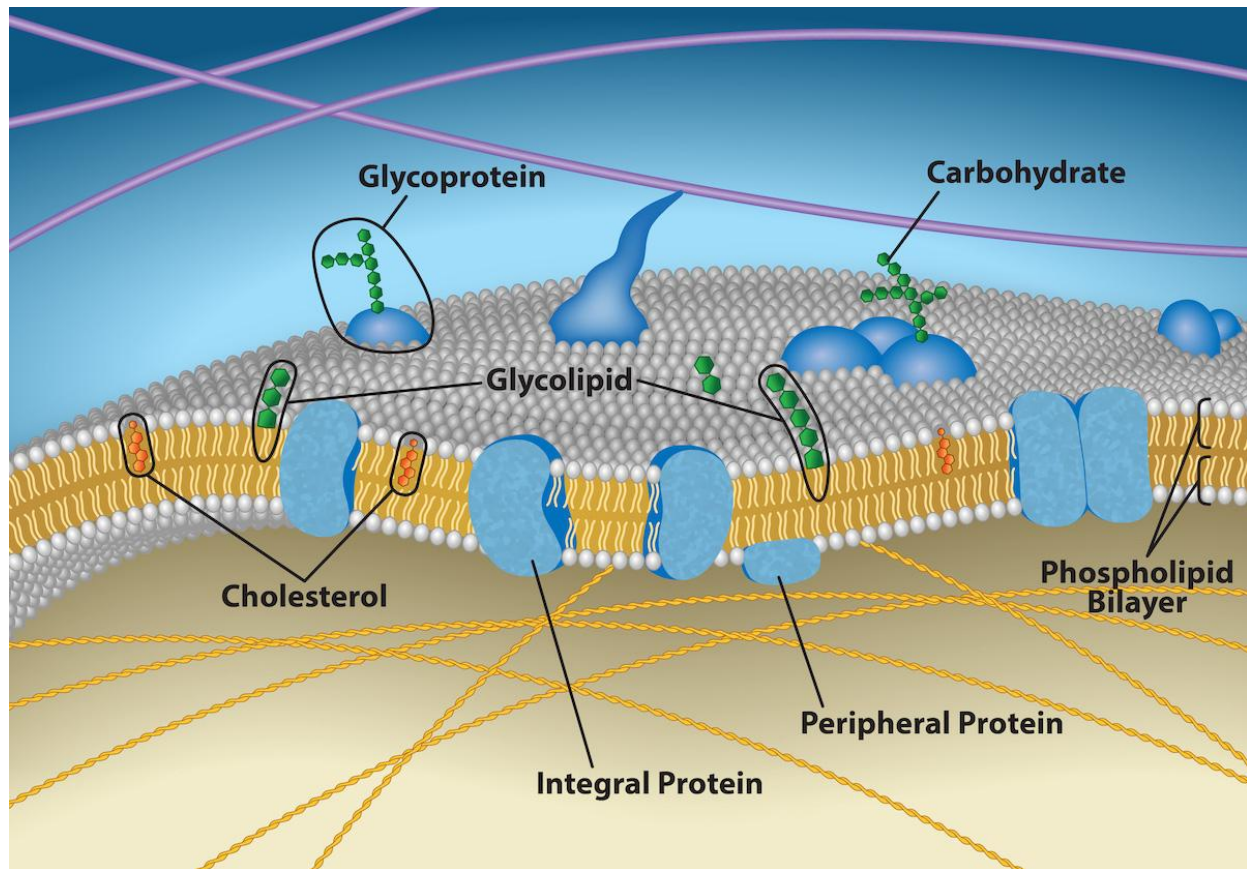
- Explain why and how passive transport occurs
- Understand the osmosis and diffusion processes
- Define tonicity and its relevance to passive transport

Plasma membranes must allow certain substances to enter and leave a cell, and prevent some harmful materials from entering and some essential materials from leaving. In other words, plasma membranes are selectively permeable (semipermeable) —they allow some substances to pass through, but not others. If they were to lose this selectivity, the cell would no longer be able to sustain itself, and it would be destroyed. Some cells require larger amounts of specific substances. They must have a way of obtaining these materials from extracellular fluids. This may happen passively, as certain materials move back and forth, or the cell may have special mechanisms that facilitate transport. Some materials are so important to a cell that it spends some of its energy, hydrolyzing adenosine triphosphate (ATP), to obtain these materials.

The most direct forms of membrane transport are passive. Passive transport is a naturally occurring phenomenon and does not require the cell to exert any of its energy to accomplish the movement. In passive transport, substances move from an area of higher concentration to an area of lower concentration.

Selective Permeability

Plasma membranes are asymmetric: the membrane's interior is not identical to its exterior. There is a considerable difference between the array of phospholipids and proteins between the two leaflets that form a membrane. On the membrane's interior, some proteins serve to anchor the membrane to cytoskeleton's fibers. There are peripheral proteins on the membrane's exterior that bind extracellular matrix elements. Carbohydrates, attached to lipids or proteins, are also on the plasma membrane's exterior surface. These carbohydrate complexes help the cell bind required substances in the extracellular fluid. This adds considerably to plasma membrane's selective nature ([Figure 5.7](#)).



This illustration shows that the inside and outside of a plasma membrane are different, with the exterior covered in the spherical heads, and the interior filled with the strandlike tails.

Figure 5.7 The plasma membrane's exterior surface is not identical to its interior surface. Credit: Rao, A., Ryan, K., Fletcher, S., Hawkins, A. and Tag, A. Department of Biology, Texas A&M University.

Recall that plasma membranes are amphipathic: They have hydrophilic and hydrophobic regions. This characteristic helps move some materials through the membrane and hinders the movement of others. Non-polar molecules can easily slip through the membrane's hydrophobic lipid core. Oxygen and carbon dioxide molecules have no charge and pass through membranes by simple passive diffusion.

Polar substances present problems for the membrane. While some polar molecules connect easily with the cell's outside, they cannot readily pass through the plasma membrane's lipid core. Additionally, while small ions could easily slip through the spaces in the membrane's mosaic, their charge prevents them from doing so. Ions such as sodium, potassium, calcium, and chloride must have special means of penetrating plasma membranes. Simple sugars and amino acids also need the help of various transmembrane proteins (channels) to transport themselves across plasma membranes.

Diffusion

Diffusion is a passive process of transport. A single substance moves from a high concentration to a low concentration area until the concentration is equal. Materials move within the cell's cytosol by diffusion, and certain materials move through the plasma membrane by diffusion ([Figure 5.8](#)). Diffusion expends no energy.

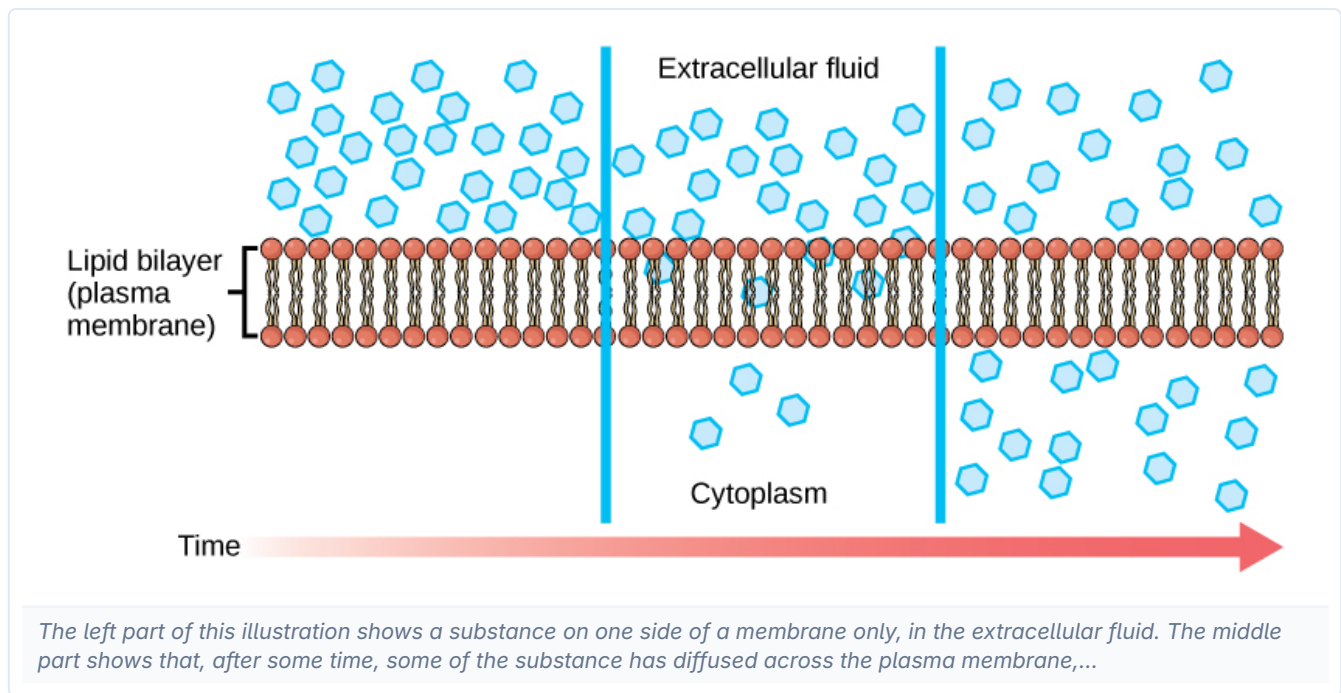


Figure 5.8 Diffusion through a permeable membrane moves a substance from a high concentration area (extracellular fluid, in this case) down its concentration gradient (into the cytoplasm). (credit: modification of work by Mariana Ruiz Villareal)

Factors That Affect Diffusion

Molecules move constantly in a random manner, at a rate that depends on their mass, their environment, and the amount of thermal energy they possess, which in turn is a function of temperature. This movement accounts for molecule diffusion through whatever medium in which they are localized. A substance moves into any space available to it until it evenly distributes itself throughout. After a substance has diffused completely through a space, removing its concentration gradient, molecules will still move around in the space, but there will be no *net* movement of the number of molecules from one area to another. We call this lack of a concentration gradient in which the substance has no net movement dynamic equilibrium. While diffusion will go forward in the presence of a substance's concentration gradient, several factors affect the diffusion rate.

- **Extent of the concentration gradient:** The greater the difference in concentration, the more rapid the diffusion. The closer the distribution of the material gets to equilibrium, the slower the diffusion rate.
- **Mass of the molecules diffusing:** Heavier molecules move more slowly; therefore, they diffuse more slowly. The reverse is true for lighter molecules.
- **Temperature:** Higher temperatures increase the energy and therefore the molecules' movement, increasing the diffusion rate. Lower temperatures decrease the molecules' energy, thus decreasing the diffusion rate.

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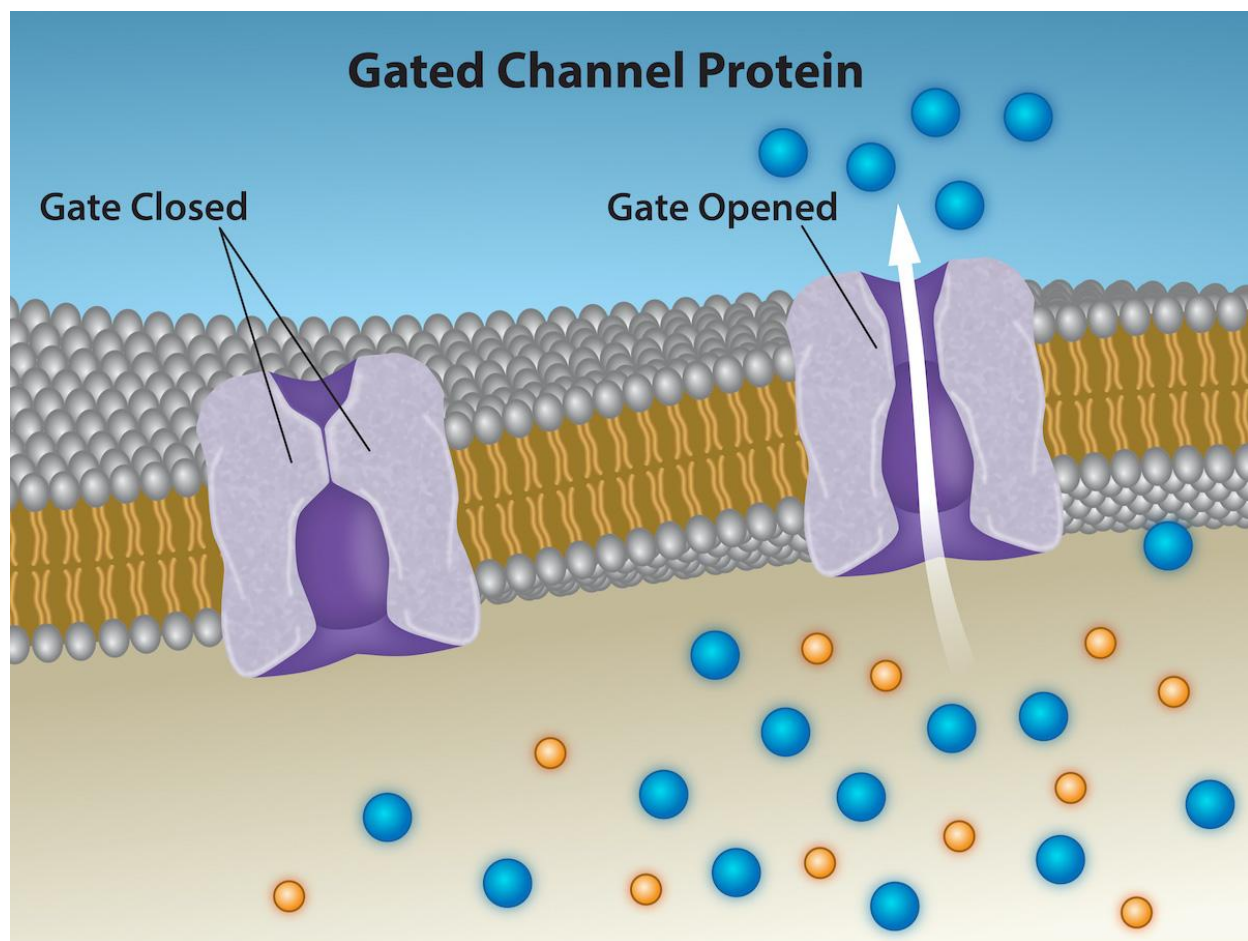
Facilitated transport

In facilitated transport, or **facilitated diffusion**, materials diffuse across the plasma membrane with the help of membrane proteins. A concentration gradient exists that would allow these materials to diffuse into the cell without expending cellular energy. However, these materials are polar molecule ions that the cell membrane's hydrophobic parts repel. Facilitated transport proteins shield these materials from the membrane's repulsive force, allowing them to diffuse into the cell.

The transported material first attaches to protein or glycoprotein receptors on the plasma membrane's exterior surface. This allows removal of material from the extracellular fluid that the cell needs. The substances then pass to specific integral proteins that facilitate their passage. Some of these integral proteins are collections of beta-pleated sheets that form a pore or channel through the phospholipid bilayer. Others are carrier proteins which bind with the substance and aid its diffusion through the membrane.

Channels

The integral proteins involved in facilitated transport are transport proteins, and they function as either **channels** for the material or **carriers**. In both cases, they are transmembrane proteins. Channels are specific for the transported substance. Channel proteins have hydrophilic domains exposed to the intracellular and extracellular fluids. In addition, they have a hydrophilic channel through their core that provides a hydrated opening through the membrane layers (Figure 5.9). Passage through the channel allows polar compounds to avoid the plasma membrane's nonpolar central layer that would otherwise slow or prevent their entry into the cell. Aquaporins are channel proteins that allow water to pass through the membrane at a very high rate.

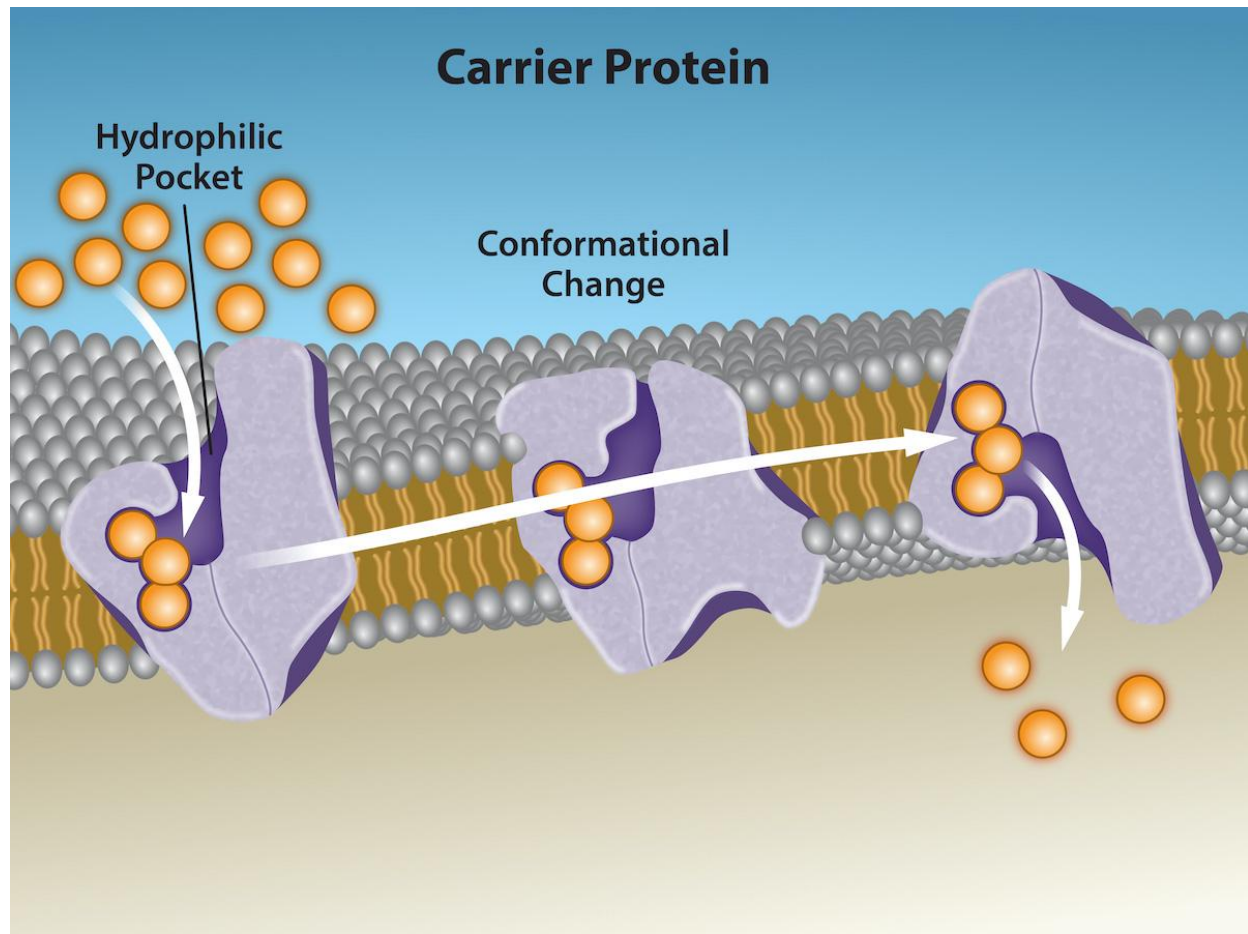


This illustration shows two channel proteins in the plasma membrane. One protein is labeled gate closed, and there is no space for particles to pass through. The other is labeled gate open, and there is a space for subs...

Figure 5.9 Ion Channel Proteins. Ion channel proteins are gated. When they are closed, no ions can pass through them. However, when a channel opens, select ions diffuse through the channel. Channel proteins are highly specific, letting only a specific ion or subset of ions pass. Credit: Rao, A., Ryan, K., Tag, A. and Fletcher, S. Department of Biology, Texas A&M University.

Carrier Proteins

Another type of protein embedded in the plasma membrane is a carrier protein. This aptly named protein binds a substance and, thus triggers a change of its own shape, moving the bound molecule from the cell's outside to its interior ([Figure 5.10](#)). Depending on the gradient, the material may move in the opposite direction. Each carrier protein is specific to one substance, and there are a finite number of these proteins in any membrane. This can cause problems in transporting enough material for the cell to function properly.



This illustration shows a carrier protein embedded in the membrane with an opening that initially faces the extracellular surface, labeled the hydrophilic pocket. After a substance binds the carrier, it changes shape so ...

Figure 5.10 Some substances are able to move down their concentration gradient across the plasma membrane with the aid of carrier proteins. Carrier proteins change shape as they move molecules across the membrane.

Credit: Rao, A., Tag, A. and Fletcher, S. Department of Biology, Texas A&M University.

Channel and carrier proteins transport material at different rates. Channel proteins transport much more quickly than carrier proteins. Channel proteins facilitate diffusion at a rate of tens of millions of molecules per second; whereas, carrier proteins work at a rate of a thousand to a million molecules per second.

Osmosis

Osmosis is the movement of free water molecules through a semipermeable membrane according to the water's concentration gradient across the membrane, which is inversely proportional to the solutes' concentration. While diffusion transports material across membranes and within cells, osmosis transports *only water* across a membrane

and the membrane limits the solutes' diffusion in the water. Not surprisingly, the aquaporins that facilitate water movement play a large role in osmosis, most prominently in red blood cells and the membranes of kidney tubules.

Mechanism

Osmosis is a special case of diffusion. Water, like other substances, moves from an area of high concentration of free water molecules to one of low free water molecule concentration. An obvious question is what makes water move at all? Imagine a beaker with a semipermeable membrane separating the two sides or halves ([Figure 5.11](#)). On both sides of the membrane the water level is the same, but there are different dissolved substance concentrations, or solute, that cannot cross the membrane (otherwise the solute crossing the membrane would balance concentrations on each side). If the solution's volume on both sides of the membrane is the same, but the solute's concentrations are different, then there are different amounts of water, the solvent, on either side of the membrane.

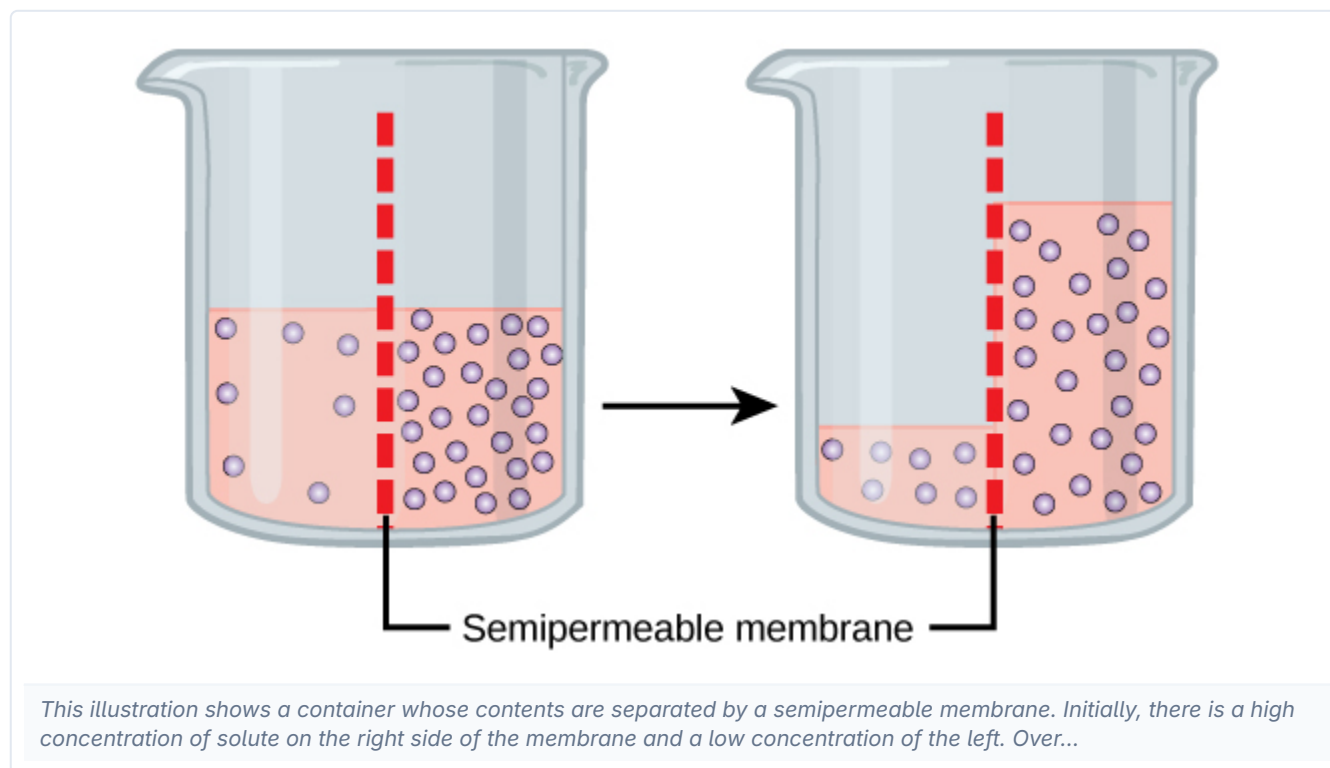


Figure 5.11 In osmosis, water always moves from an area of higher water concentration to one of lower concentration. In the diagram, the solute cannot pass through the selectively permeable membrane, but the water can.

To illustrate this, imagine two full water glasses. One has a single teaspoon of sugar in it; whereas, the second one contains one-quarter cup of sugar. If the total volume of the solutions in both cups is the same, which cup contains more water? Because the large sugar amount in the second cup takes up much more space than the teaspoon of sugar in the first cup, the first cup has more water in it.

Returning to the beaker example, recall that it has a solute mixture on either side of the membrane. A principle of diffusion is that the molecules move around and will spread evenly throughout the medium if they can. However, only the material capable of getting through the membrane will diffuse through it. In this example, the solute cannot diffuse through the membrane, but the water can. Water has a concentration gradient in this system. Thus, water will diffuse down its concentration gradient, crossing the membrane to the side where it is less concentrated. This diffusion of water through the membrane—osmosis—will continue until the water's concentration gradient goes to

zero or until the water's hydrostatic pressure balances the osmotic pressure. Osmosis proceeds constantly in living systems.

Tonicity

Tonicity describes how an extracellular solution can change a cell's volume by affecting osmosis. In a situation in which a membrane permeable to water, though not to the solute, water will move from the membrane's side with higher water concentration to the side with lower water concentration. This effect makes sense if you remember that the solute cannot move across the membrane, and thus the only component in the system that can move—the water—moves along its own concentration gradient.

Hypotonic Solutions

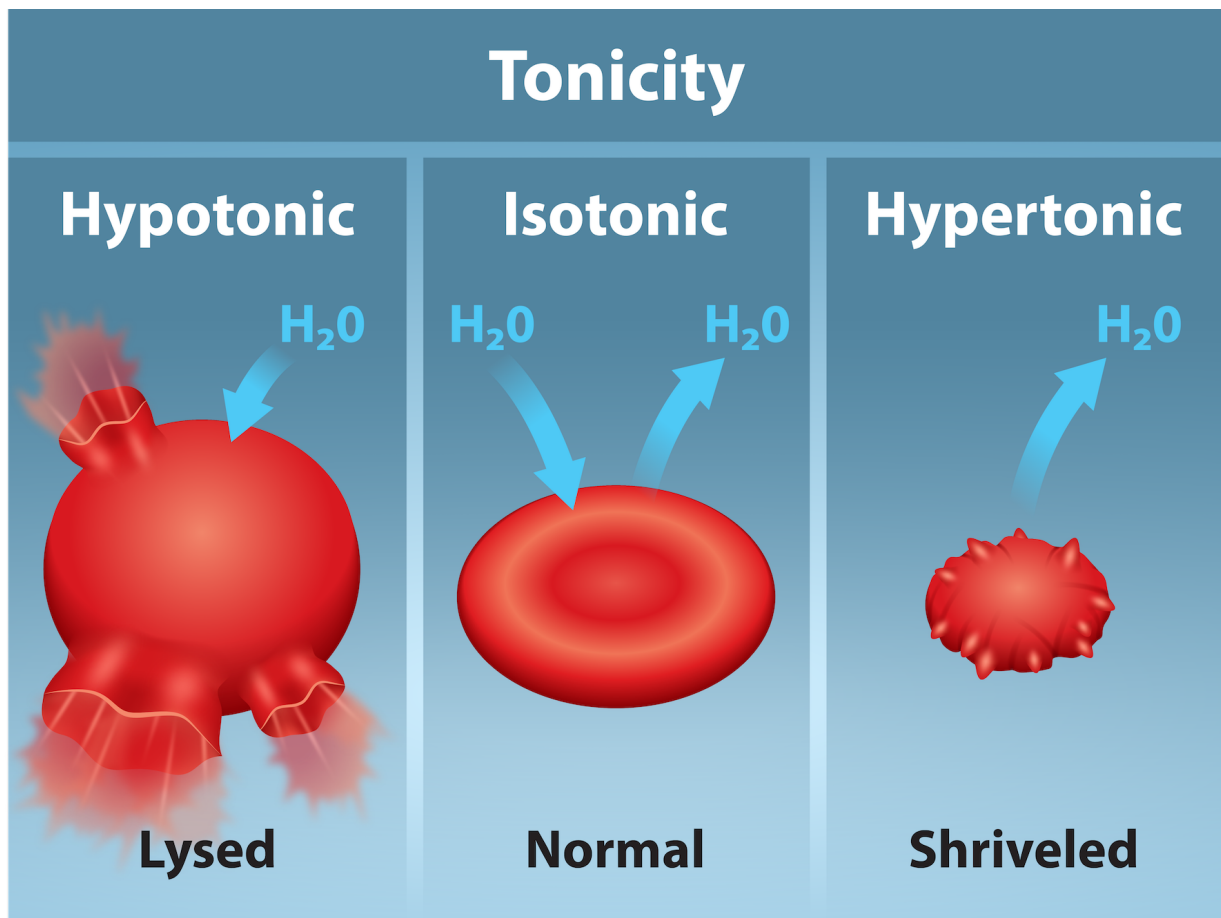
Scientists use three terms—hypotonic, isotonic, and hypertonic—to relate the cell's osmolarity to the extracellular fluid's osmolarity that contains the cells. In a hypotonic situation, the extracellular fluid has a greater concentration of water than inside the cell, and water will follow its concentration gradient and enter the cell.

Hypertonic Solutions

As for a hypertonic solution, the extracellular fluid has a lower water concentration than the cell's cytoplasm. Because the cell has a relatively higher water concentration, water will leave the cell.

Isotonic Solutions

In an isotonic solution, the extracellular fluid has the water concentration as the cell. In this situation there will be no net movement of water into or out of the cell, although water will still move in and out equally. Blood cells and plant cells in hypertonic, isotonic, and hypotonic solutions take on characteristic appearances ([Figure 5.12](#)).



The left part of this illustration shows shriveled red blood cells bathed in a hypertonic solution. The middle part shows healthy red blood cells bathed in an isotonic solution, and the right part shows bloated red blood...

Figure 5.12 Three different scenarios involving red blood cells (RBC) are shown. Left: A RBC placed in a hypotonic solution, where the concentration of solutes in the surrounding fluid is lower than those in the cell, will cause water to rush into the RBC and lead to lysis of the cell. Middle: there is no net water movement into or out of the cell as the concentration of the solutes inside the cell equal or is isotonic to that of the surrounding fluid. Right: a RBC placed in a hypertonic solution, where the concentration of solutes in the surrounding fluid is greater than that in the cell will cause water to rush out of the cell and into the surrounding fluid. This will cause the RBC to shrivel. Credit: Tag, A., Rao, A., Hawkins, A and Fletcher, S. Department of Biology, Texas A&M University.

Tonicity in Living Systems

In a hypotonic environment, water enters a cell, and the cell swells. In an isotonic condition, the relative solute and solvent concentrations are equal on both membrane sides. There is no net water movement; therefore, there is no change in the cell's size. In a hypertonic solution, water leaves a cell and the cell shrinks. If either the hypo- or hyper- condition goes to excess, the cell's functions become compromised, and the cell may be destroyed.

A red blood cell will burst, or **lyse**, when it swells beyond the plasma membrane's capability to expand. Remember, the membrane resembles a mosaic, with discrete spaces between the molecules comprising it. If the cell swells, and the spaces between the lipids and proteins become too large, the cell will break apart.

In contrast, when excessive water amounts leave a red blood cell, the cell shrinks, or **crenates**. This has the effect of concentrating the solutes left in the cell, making the cytosol denser and interfering with diffusion within the cell. The cell's ability to function will be compromised and may also result in the cell's death.

Module 3: Chapter 7 Active Transport

Source: Chapter 7 Active Transport_ BSC2010C-26Fall 0001.html

Chapter 7 Active Transport

Learning Objectives

By the end of this section, you will be able to do the following:

- Understand how electrochemical gradients affect ions
- Distinguish between primary active transport and secondary active transport
- Describe endocytosis, including phagocytosis, pinocytosis, and receptor-mediated endocytosis
- Understand the process of exocytosis

Active transport mechanisms require the cell's energy, usually in the form of adenosine triphosphate (ATP). If a substance must move into the cell against its concentration gradient—that is, if the substance's concentration inside the cell is greater than its concentration in the extracellular fluid (and vice versa)—the cell must use energy to move the substance. Some active transport mechanisms move small-molecular weight materials, such as ions, through the membrane. Other mechanisms transport much larger molecules.

Moving Against a Gradient

To move substances against a concentration or electrochemical gradient, the cell must use energy. This energy comes from ATP generated through the cell's metabolism. Active transport mechanisms, or pumps, work against electrochemical gradients. Small substances constantly pass through plasma membranes. Active transport maintains concentrations of ions and other substances that living cells require in the face of these passive movements. A cell may spend much of its metabolic energy supply maintaining these processes. (A red blood cell uses most of its metabolic energy to maintain the imbalance between exterior and interior sodium and potassium levels that the cell requires.) Because active transport mechanisms depend on a cell's metabolism for energy, they are sensitive to many metabolic poisons that interfere with the ATP supply.

Active transport moves ions, charged molecules, and large molecules across a membrane. Active transport requires energy in the form of ATP.

Carrier Proteins for Active Transport

An important membrane adaptation for active transport is the presence of specific carrier proteins or pumps to facilitate movement: there are three protein types or transporters ([Figure 5.18](#)). A uniporter carries one specific ion or molecule. A symporter carries two different ions or molecules, both in the same direction. An antiporter also carries two different ions or molecules, but in different directions. All of these transporters can also transport small, uncharged organic molecules like glucose. These three types of carrier proteins are also in facilitated passive diffusion, but they do not require ATP to work in that process. Some examples of pumps for active transport are $\text{Na}^+\text{-K}^+$ ATPase, which carries sodium and potassium ions, and $\text{H}^+\text{-K}^+$ ATPase, which carries hydrogen and potassium ions. Both of these are antiporter carrier proteins. Two other carrier proteins are Ca^{2+} ATPase and H^+ ATPase, which carry only calcium and only hydrogen ions, respectively. Both are pumps.

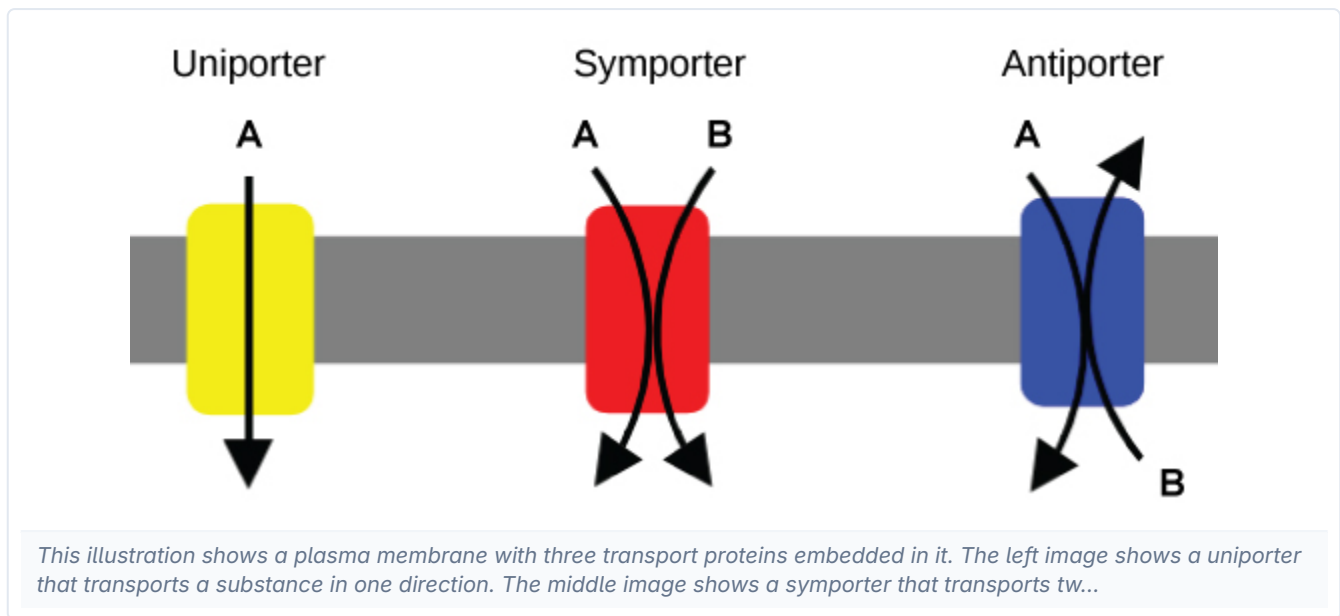


Figure 5.18 A uniporter carries one molecule or ion. A symporter carries two different molecules or ions, both in the same direction. An antiporter also carries two different molecules or ions, but in different directions. (credit: modification of work by "Lupask"/Wikimedia Commons)

Active Transport

The primary active transport that functions with the active transport of sodium and potassium allows secondary active transport to occur.

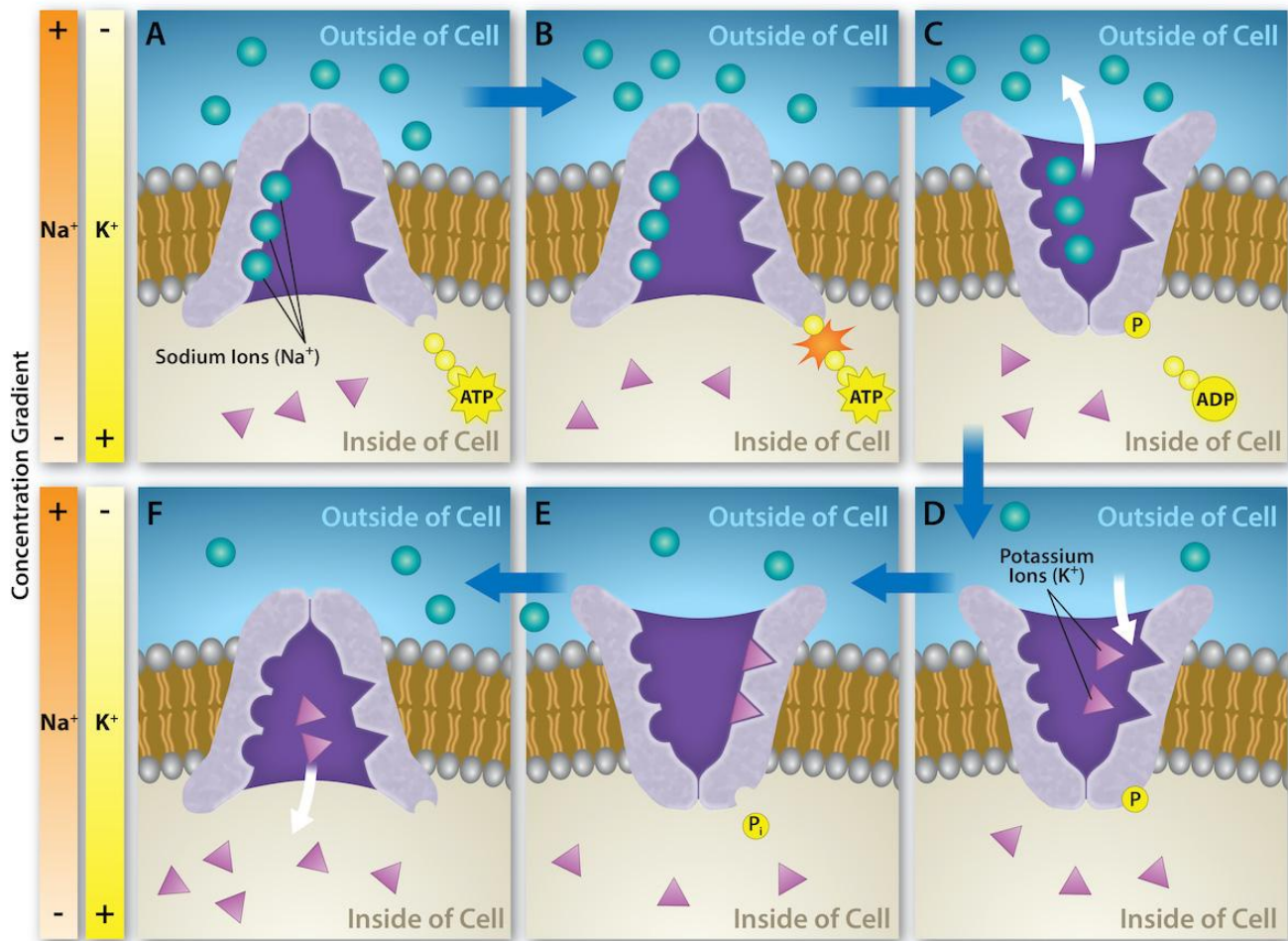


Figure 5.19 The sodium-potassium pump is an example of primary active transport that moves ions, sodium and potassium ions in this instance, across a membrane against their concentration gradients. The energy is provided by the hydrolysis of ATP. Three sodium ions are moved out of the cell for every 2 potassium ions that are brought into the cell. Credit: Rao, A., Ryan, K. and Fletcher, S. Department of Biology, Texas A&M University.

One of the most important pumps in animal cells is the sodium-potassium pump (Na^+/K^+ ATPase), which maintains the electrochemical gradient (and the correct concentrations of Na^+ and K^+) in living cells. The sodium-potassium pump moves K^+ into the cell while moving Na^+ out at the same time, at a ratio of three Na^+ for every two K^+ ions moved in. The Na^+/K^+ ATPase exists in two forms, depending on its orientation to the cell's interior or exterior and its affinity for either sodium or potassium ions. The process consists of the following six steps.

1. With the enzyme oriented towards the cell's interior, the carrier has a high affinity for sodium ions. Three ions bind to the protein.
2. The protein carrier hydrolyzes ATP and a low-energy phosphate group attaches to it.
3. As a result, the carrier changes shape and reorients itself towards the membrane's exterior. The protein's affinity for sodium decreases and the three sodium ions leave the carrier.
4. The shape change increases the carrier's affinity for potassium ions, and two such ions attach to the protein. Subsequently, the low-energy phosphate group detaches from the carrier.
5. With the phosphate group removed and potassium ions attached, the carrier protein repositions itself towards the cell's interior.

6. The carrier protein, in its new configuration, has a decreased affinity for potassium, and the two ions move into the cytoplasm. The protein now has a higher affinity for sodium ions, and the process starts again.

Module 4: Chapter 7 Bulk Transport

Source: Chapter 7 Bulk Transport_ BSC2010C-26Fall 0001.html

Chapter 7 Bulk Transport

Learning Objectives

By the end of this section, you will be able to do the following:

- Describe endocytosis, including phagocytosis, pinocytosis, and receptor-mediated endocytosis
- Understand the process of exocytosis

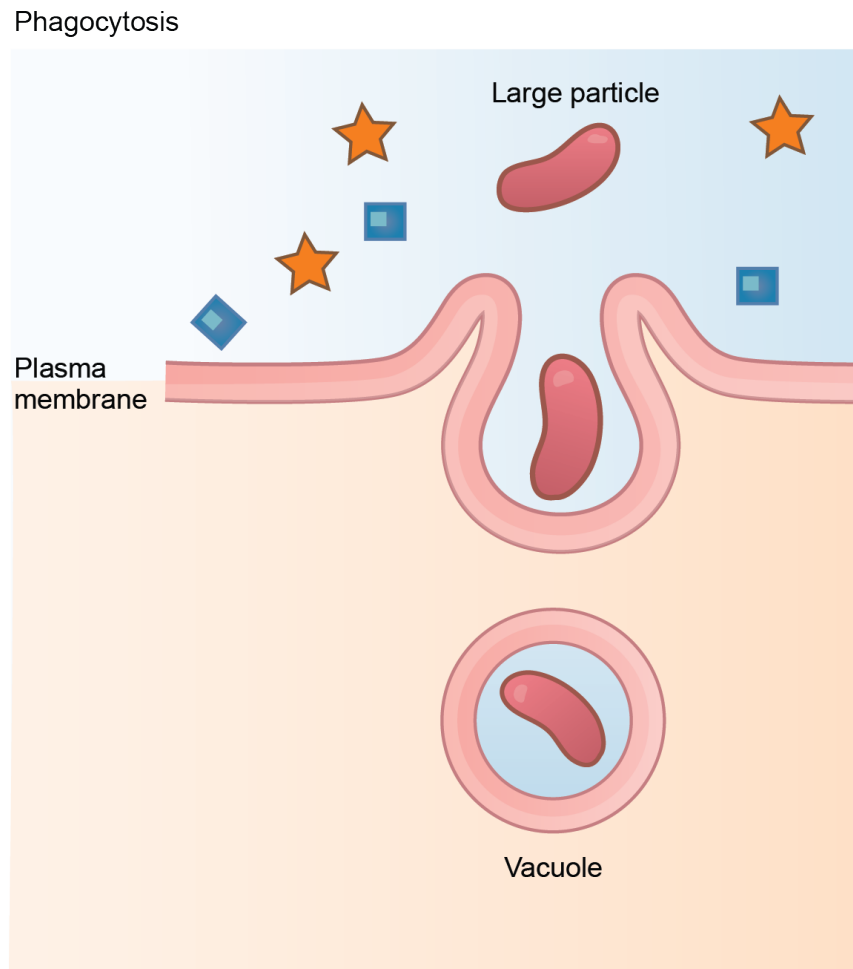
In addition to moving small ions and molecules through the membrane, cells also need to remove and take in larger molecules and particles (see [Table 5.2](#) for examples). Some cells are even capable of engulfing entire unicellular microorganisms.

Endocytosis

Endocytosis is a type of active transport that moves particles, such as large molecules, parts of cells, and even whole cells, into a cell. There are different endocytosis variations, but all share a common characteristic: the cell's plasma membrane invaginates, forming a pocket around the target particle. The pocket pinches off, resulting in the particle containing itself in a newly created intracellular vesicle formed from the plasma membrane.

Phagocytosis

Phagocytosis (the condition of “cell eating”) is the process by which a cell takes in large particles, such as other cells or relatively large particles. For example, when microorganisms invade the human body, a type of white blood cell, a neutrophil, will remove the invaders through this process, surrounding and engulfing the microorganism, which the neutrophil then destroys ([Figure 5.21](#)).



This illustration shows a plasma membrane forming a pocket around a particle in the extracellular fluid. The membrane subsequently engulfs the particle, which becomes trapped in a vacuole.

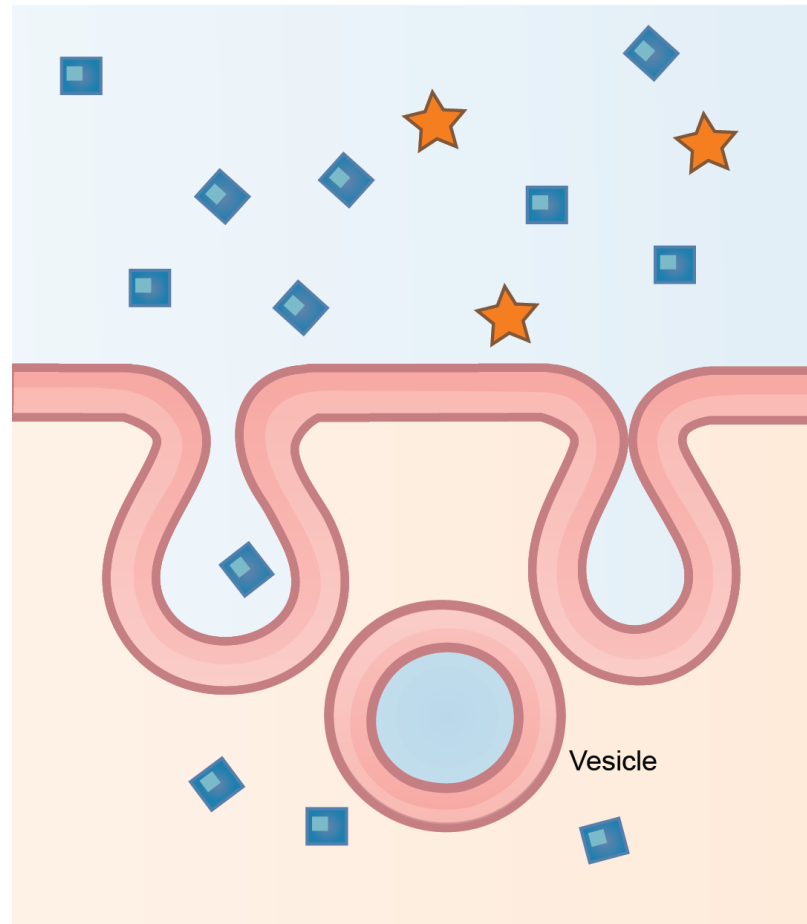
Figure 5.21 In phagocytosis, the cell membrane surrounds the particle and engulfs it. (credit: modification of work by Mariana Ruiz Villareal)

In preparation for phagocytosis, a portion of the plasma membrane's inward-facing surface and then extends from the cell's body and surrounds the particle, eventually enclosing it. Once the vesicle containing the particle is enclosed within the cell, the vesicle merges with a lysosome for breaking down the material in the newly formed compartment (endosome).

Pinocytosis

A variation of endocytosis is pinocytosis. This literally means "cell drinking". Discovered by Warren Lewis in 1929, this American embryologist and cell biologist described a process whereby he assumed that the cell was purposefully taking in extracellular fluid. In reality, this is a process that takes in molecules, including water, which the cell needs from the extracellular fluid. Pinocytosis results in a much smaller vesicle than does phagocytosis, and the vesicle does not need to merge with a lysosome ([Figure 5.22](#)).

Pinocytosis



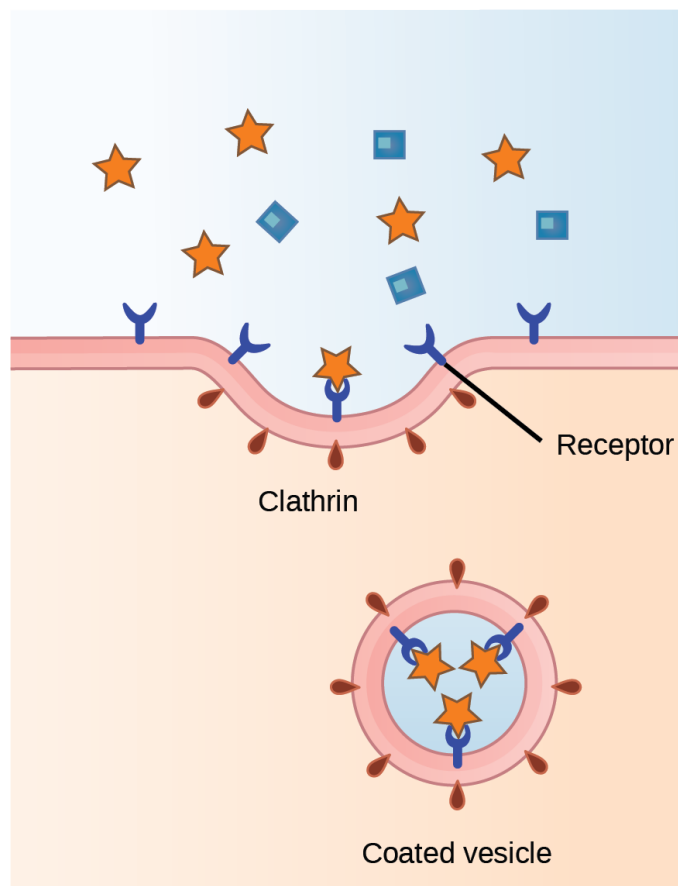
This illustration shows a plasma membrane forming a pocket around fluid in the extracellular fluid. The membrane subsequently engulfs the fluid, which becomes trapped in a vacuole.

Figure 5.22 In pinocytosis, the cell membrane invaginates, surrounds a small volume of fluid, and pinches off. (credit: modification of work by Mariana Ruiz Villareal)

Receptor-mediated Endocytosis

A targeted variation of endocytosis employs receptor proteins in the plasma membrane that have a specific binding affinity for certain substances ([Figure 5.23](#)). Although receptor-mediated endocytosis is designed to bring specific substances that are normally in the extracellular fluid into the cell, other substances may gain entry into the cell at the same site. Flu viruses, diphtheria, and cholera toxin all have sites that cross-react with normal receptor-binding sites and gain entry into cells.

Receptor-mediated endocytosis



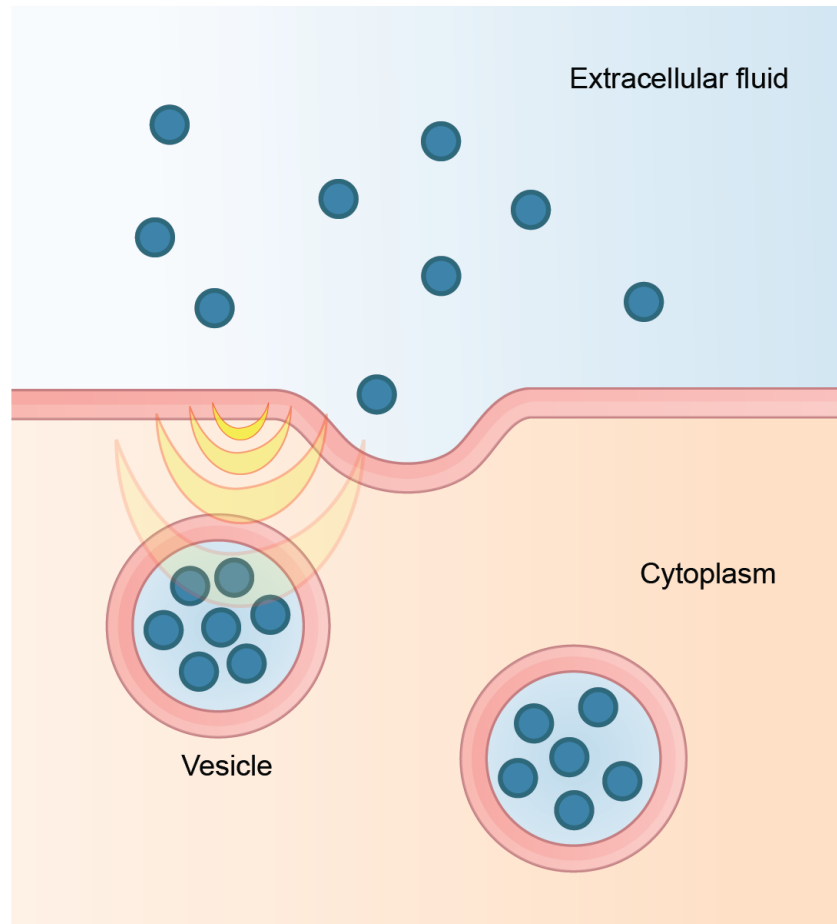
This illustration shows a part of the plasma membrane that is clathrin-coated on the cytoplasmic side and has receptors on the extracellular side. The receptors bind a substance, then pinch off to form a vesicle.

Figure 5.23 In receptor-mediated endocytosis, the cell's uptake of substances targets a single type of substance that binds to the receptor on the cell membrane's external surface. (credit: modification of work by Mariana Ruiz Villareal)

Exocytosis

The reverse process of moving material into a cell is the process of exocytosis. Exocytosis is the opposite of the processes we discussed above in that its purpose is to expel material from the cell into the extracellular fluid. Waste material is enveloped in a membrane and fuses with the plasma membrane's interior. This fusion opens the membranous envelope on the cell's exterior, and the waste material expels into the extracellular space ([Figure 5.24](#)).

Exocytosis



This illustration shows vesicles fusing with the plasma membrane and releasing their contents to the extracellular fluid.

Figure 5.24 In exocytosis, vesicles containing substances fuse with the plasma membrane. The contents then release to the cell's exterior. (credit: modification of work by Mariana Ruiz Villareal)

Appendix: Original Course Lecture Practice Problems & Solutions

Source: Chapter 7 Student Lecture Presentations (Parts 1 & 2)

Part A: Membrane Permeability Check (Lecture Part 1, Slide 22)

Question: For each of the following molecules, determine whether a transport protein is required to cross the plasma membrane, or if it crosses via simple diffusion:

- **Water (H_2O):** Requires transport protein (Aquaporin) for rapid physiological volume flow, though very small amounts can slowly cross the bilayer.
- **Oxygen (O_2):** No transport protein needed. Small, nonpolar diatomic gas diffuses rapidly across the hydrophobic core down its concentration gradient.
- **Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$):** Requires transport protein (GLUT carrier protein). Large uncharged polar hexose cannot penetrate the hydrophobic interior.
- **Carbon Dioxide (CO_2):** No transport protein needed. Small, nonpolar waste gas diffuses freely down its gradient into blood capillaries.
- **Protein (Macromolecule):** Cannot cross via channel/carrier proteins. Requires vesicular bulk transport (endocytosis or exocytosis).

Part B: Tonicity Practice Problem 1 (Lecture Part 2, Slide 33)

Setup: A cell is suspended in an aqueous beaker solution under the following parameters:

- **Beaker Solution:** 20% solute • 80% water
- **Cell Interior:** 10% solute • 90% water

Analysis & Solution:

- The cell is **hypotonic** compared to the solution (lower solute, higher free water).
- The solution is **hypertonic** compared to the cell (higher solute, lower free water).
- **Direction of Osmosis:** Water diffuses **OUT of the cell** into the solution (down water's gradient: 90% → 80%). The cell will shrink / crenate.

Part B: Tonicity Practice Problem 2 (Lecture Part 2, Slide 34)

Setup: A cell is suspended in an aqueous beaker solution under the following parameters:

- **Beaker Solution:** 2% solute • 98% water
- **Cell Interior:** 2% solute • 98% water

Analysis & Solution:

- The cell is **isotonic**. The solution is **isotonic**.
- **Direction of Osmosis:** Water diffuses into and out of the cell at **equal rates** (dynamic equilibrium; no net movement). Cell volume remains constant.

Part B: Tonicity Practice Problem 3 (Lecture Part 2, Slide 35)

Setup: A cell is suspended in an aqueous beaker solution under the following parameters:

- **Beaker Solution:** 8% solute • **92% water**
- **Cell Interior:** 10% solute • **90% water**

Analysis & Solution:

- The cell is **hypertonic** compared to the solution (higher solute, lower free water).
- The solution is **hypotonic** compared to the cell (lower solute, higher free water).
- **Direction of Osmosis:** Water diffuses **INTO the cell** from the solution (92% → 90%). The cell will swell (turgid in plants, potentially lysed in animal cells).

Part B: Tonicity Practice Problem 4 (Lecture Part 2, Slide 36)

Setup: A cell with 10% solute and 90% water is placed in a solution that is 15% solute and 85% water.

Analysis & Solution:

- Is the cell hyper- or hypotonic compared to the solution? **The cell is hypotonic** (10% < 15%).
- Will water diffuse into or out of the cell? Water diffuses **OUT of the cell** (from 90% free water to 85% free water).

Part B: Tonicity Practice Problem 5 (Lecture Part 2, Slide 37)

Setup: A red blood cell (~1% solute, 99% water) is placed in a solution that is 5% solute and 95% water.

Analysis & Solution:

- Is the red blood cell iso-, hyper-, or hypotonic compared to the solution? **The red blood cell is hypotonic** relative to the hypertonic 5% solution.
- Will water diffuse into or out of the cell? Water diffuses **OUT of the cell** via osmosis.
- What happens to the cell? The red blood cell loses water, shrivels, and undergoes **crenation**, impairing its oxygen-carrying capacity.

Part B: Tonicity Practice Problem 6 (Lecture Part 2, Slide 38)

Setup: A stalk of celery (~1% solute) is starting to wilt. To perk it up, would you place the celery in: A. pure water, B. a solution that is 5% sugar, C. a solution that is 10% sugar? Why?

Analysis & Solution:

- **Correct Choice: A. Pure water (0% solute / 100% water).**
- **Biological Rationale:** Pure water is strongly **hypotonic** relative to the celery cells' cytoplasm (1% solute). Water spontaneously rushes into the plant cells via osmosis, filling the central vacuoles. The expanding vacuoles exert high hydrostatic **turgor pressure** against the rigid cellulose cell walls, reviving the crisp, erect structural rigidity of the celery stalk. Placing it in 5% or 10% sugar (hypertonic solutions) would draw water out, causing plasmolysis and worsening the wilt!